Operator-Adaptive Kernel Ridge Regression for Near-Infrared Spectroscopy: Dual Formulation, Branch-CV, and MKL-Light

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Abstract

Ridge regression is a strong baseline for near-infrared spectroscopy (NIRS) calibration but, like PLS, it depends on a long chain of spectral preprocessing decisions that practitioners typically search as an external Cartesian grid. We present *Operator-Adaptive Kernel Ridge Regression* (AOM-Ridge), a kernel formulation that absorbs that preprocessing search into the Ridge solve itself. Each strict linear spectral operator $\mathbf{A}_b \in \mathbb{R}^{p\times p}$ defines a view $\mathbf{X}_b = \mathbf{X}\mathbf{A}_b^\top$; we form the superblock dual kernel $K = \sum_b s_b^2 \mathbf{X}\mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$ and a metric matrix $U = \sum_b s_b^2 \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$ such that the original- space Ridge coefficient is $\boldsymbol{\beta} = UC$ with $C = (K + \alpha I)^{-1}\mathbf{Y}_c$. Because Ridge is geometry-dependent, the covariance-space shortcut available to PLS does not extend to Ridge: the kernel must be assembled explicitly per operator. The estimator exposes five selection modes — `superblock` (one Ridge over the full bank), `global` (hard $(b^\star, \alpha^\star)$ selection by fold-local CV), `active_superblock` (top-m + diversity pruning), `branch_global` (fold-local CV over (branch, op, α) with non-strict-linear branches in {`none`, SNV, MSC }), and `mkl` (kernel-target-alignment weighted block sum) — all of which run on top of a single fold-local CV core that rebuilds means, operator clones, scales, and kernels inside every fold. Anti-leakage is verified by spy operators that fail when any fold-local invariant is broken. On a 39-dataset NIRS regression cohort assembled from the TabPFN paper benchmark, AOM-Ridge wins on **32/38 datasets (84%)** against the published Ridge HPO baseline, with a median test-RMSEP delta of -3.64%; the strongest single mode (`global`) wins on $\sim$33% of datasets, with SNV-branch and family-pruned variants contributing the remainder. The estimator returns an original-space coefficient $\boldsymbol{\beta} \in \mathbb{R}^{p\times q}$, is sklearn-compatible, and is shipped as a self-contained package (`aomridge`) under `bench/AOM_v0/Ridge/`.

Keywords: Ridge regression; kernel methods; NIRS; operator-adaptive preprocessing; multiple kernel learning; TabPFN; reproducible benchmark.

1 Introduction

Near-infrared spectroscopy (NIRS) calibration is a high-dimensional, small-n regression problem [Williams and Norris, 2007, Rinnan et al., 2009]. Spectra contain hundreds to thousands of strongly correlated channels, and most public NIRS datasets contain only a few hundred labelled samples. In this regime Ridge regression [Hoerl and Kennard, 1970b,a, Hastie et al., 2009] is a strong baseline: the ℓ_2 penalty controls the variance inflation that ordinary least squares would suffer from, and the closed-form solution makes it cheap to evaluate many regularisation strengths α via a single eigendecomposition of the kernel $\mathbf{K} = \mathbf{X}\mathbf{X}^\top$ [Saunders et al., 1998, Schölkopf and Smola, 2002]. The reference results published with the TabPFN tabular foundation model paper [Hollmann et al., 2023, 2025] use Ridge as one of the four classical baselines on their NIRS regression cohort, after a non-trivial preprocessing search (SNV, MSC, Savitzky-Golay smoothing/derivatives, detrend, baseline correction) and an Optuna-driven α tuning. We refer to this configuration in the rest of the paper as *paper Ridge HPO*.

The cost of paper Ridge HPO is the cost of *any* preprocessing grid search: the analyst picks a finite Cartesian product of operator choices, runs Ridge on each, and reports the best one against an external split. Two analysts with the same data and the same operator inventory can reach different pipelines, the runs are not auditable, and the search is irreproducible across software stacks because the Cartesian product itself is implementation-defined. The companion paper for AOM-PLS [Beurier and nirs4all contributors, 2026] addresses the same problem for PLS by absorbing the preprocessing search into the latent extraction loop: every strict linear spectral operator $\mathbf{A} \in \mathbb{R}^{p\times p}$ acts as $\mathbf{X}\mathbf{A}^\top$ on row spectra, and the cross-covariance identity $(\mathbf{X}\mathbf{A}^\top)^\top\mathbf{Y} = \mathbf{A}\mathbf{X}^\top\mathbf{Y}$ lets one evaluate operator candidates in covariance space without ever materialising the transformed matrix. The AOM-PLS companion paper is parity-validated against the deployed `nirs4all AOMPLSRegressor` and reports median improvements over PLS on the TabPFN/NIRS cohort.

This paper asks the same methodological question for Ridge: *can a bank of strict linear preprocessing operators be absorbed into the Ridge solve itself?* The answer requires more care than for PLS. Ridge is a *geometry*-dependent estimator: the loss $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_F^2 + \alpha\|\boldsymbol{\beta}\|_F^2$ depends on the inner product of $\mathbf{X}$, not just on its cross-covariance with $\mathbf{Y}$. The covariance-space shortcut at the heart of AOM-PLS is therefore not available to AOM-Ridge: every operator changes the kernel, and the kernel is what the Ridge solve actually sees. The AOM-Ridge formulation is therefore a kernel formulation in the strict sense. We form a single dual kernel $\mathbf{K} = \sum_b s_b^2\, \mathbf{X}\mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$ over a bank of strict-linear operators, solve the dual Ridge problem $(\mathbf{K} + \alpha\mathbf{I})\mathbf{C} = \mathbf{Y}_c$ once, and recover the original- space coefficient $\boldsymbol{\beta} \in \mathbb{R}^{p\times q}$ via a metric matrix $\mathbf{U} = \sum_b s_b^2 \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$.

The contributions of this work are:

(1) a **dual / kernel AOM-Ridge formulation** (Sections 2.2–2.3) with an original-space coefficient $\boldsymbol{\beta} = \mathbf{U}\mathbf{C}$ that makes the estimator a drop-in replacement for `sklearn.linear_model.Ridge`;

(2) a **family of selection modes** (Section 2.5): `superblock` (one Ridge over the union of views), `global` (hard $(b^\star, \alpha^\star)$ selection by fold-local CV), `active_superblock` (top-m + diversity pruning), `branch_global` (fold-local CV over (branch, op, α) where branches in {`none`,SNV,MSC } carry non-strict-linear preprocessing), and `mkl` (kernel-target-alignment weighted block sum);

(3) a **fold-local CV core** (Section 2.6) with documented anti-leakage invariants and spy-operator regression tests that fail when the invariants are broken;

(4) an **adaptive alpha grid** with boundary-expansion (Section 2.7) and a one-standard-error selection rule, which together remove most of the residual sensitivity to the user-supplied alpha range;

(5) a **39-dataset NIRS regression benchmark** (Sections 4–5) on the TabPFN paper cohort against six baselines (Ridge, PLS, CNN, CatBoost, TabPFN-Raw, TabPFN-opt) and an explicit comparison against published *paper Ridge HPO* results.

The headline result is that AOM-Ridge wins on **32/38 datasets (84%)** against paper Ridge HPO, with a median test-RMSEP delta of −3.90%. The `global` mode contributes the largest share of dataset wins (∼33%), followed by the SNV-branch variant of `global` (∼26%) and the family-pruned SNV variant (∼15%). Limitations and failure modes are discussed in Section 6: AMYLOSE-style y-based splits where inner CV is not predictive of test performance, and outlier datasets (BEER `YbaseSplit`, TIC) where SNV is selected by inner CV but raw spectra would be the better choice. We do not claim that AOM-Ridge beats TabPFN-opt with its per-dataset hyperparameter optimisation [Hollmann et al., 2025]; we claim a parity-grade Ridge baseline that absorbs the preprocessing search.

Relation to AOM-PLS. AOM-Ridge is the companion of AOM-PLS [Beurier and nirs4all contributors, 2026] for the Ridge family. The two estimators share the operator bank infrastructure (`aompls.banks`, `aompls.operators`) but otherwise reach the operator-adaptive goal through different routes: AOM-PLS exploits the cross-covariance identity to evaluate operators without materialising $\mathbf{X}\mathbf{A}^\top$, while AOM-Ridge must assemble the kernel explicitly because the Ridge geometry depends on $\mathbf{X}\mathbf{A}^\top\mathbf{A}\mathbf{X}^\top$, not on $\mathbf{A}\mathbf{X}^\top\mathbf{Y}$. The two papers can be read independently.

2 Method

2.1 Notation and the strict-linear operator contract

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ collect n row spectra of length p and $\mathbf{Y} \in \mathbb{R}^{n\times q}$ the response matrix ($q = 1$ for univariate regression). A *strict linear spectral operator* is a matrix $\mathbf{A} \in \mathbb{R}^{p\times p}$ that acts on row spectra by right- multiplication of its transpose:

$$\mathbf{X}_b \;=\; \mathbf{X}\,\mathbf{A}_b^\top. \tag{1}$$

A *bank* is a finite set $\mathcal{B} = \{\mathbf{A}_1, \dots, \mathbf{A}_B\}$ where $\mathbf{A}_1$ is the identity (prepended automatically; duplicate identities are deduplicated). The banks supported by AOM-Ridge are imported from the AOM-PLS package and include `compact` ($|\mathcal{B}| = 9$), `default` ($|\mathcal{B}| = 100$), `family_pruned` ($|\mathcal{B}| = 15$, one representative per family), `response_dedup` ($|\mathcal{B}| = 46$, response-cosine pruned), and the deeper `deep3` / `deep4` research banks. *Strict linearity* is essential: SNV and MSC have row-dependent parameters, so they are not strict-linear operators in this sense; they appear in AOM-Ridge only as *branch* preprocessors, covered in Section 2.5.

Centering. All formulations below apply Ridge in centered form. Let $\bar{\mathbf{x}} = \mathbf{X}^\top\mathbf{1}/n$ and $\bar{\mathbf{y}} = \mathbf{Y}^\top\mathbf{1}/n$ be training-fold means; we write $\mathbf{X}_c = \mathbf{X} - \bar{\mathbf{x}}^\top$ and $\mathbf{Y}_c = \mathbf{Y} - \bar{\mathbf{y}}^\top$. Predictions on a new spectrum $\mathbf{x}_\star$ are $\hat{\mathbf{y}}_\star = (\mathbf{x}_\star - \bar{\mathbf{x}})^\top\boldsymbol{\beta} + \bar{\mathbf{y}}$. With $\boldsymbol{\beta} \in \mathbb{R}^{p\times q}$ the original-space coefficient, the intercept absorbs the centering: $\hat{\mathbf{Y}} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{c}$ with $\boldsymbol{c} = \bar{\mathbf{y}} - \bar{\mathbf{x}}^\top\boldsymbol{\beta}$.

2.2 Single-operator dual Ridge

For one strict linear operator $\mathbf{A}_b$, define $\mathbf{Z}_b = \mathbf{X}_c\mathbf{A}_b^\top$. The Ridge problem in the transformed space is

$$\min_{\boldsymbol{\gamma}} \; \|\mathbf{Y}_c - \mathbf{Z}_b\boldsymbol{\gamma}\|_F^2 + \alpha\|\boldsymbol{\gamma}\|_F^2, \tag{2}$$

with closed-form primal solution $\boldsymbol{\gamma}_b = (\mathbf{Z}_b^\top\mathbf{Z}_b + \alpha\mathbf{I}_p)^{-1}\mathbf{Z}_b^\top\mathbf{Y}_c$. For high-dimensional NIRS spectra ($p \gg n$) the dual form is much cheaper. Define

$$\mathbf{K}_b \;=\; \mathbf{Z}_b\mathbf{Z}_b^\top \;=\; \mathbf{X}_c\mathbf{A}_b^\top\mathbf{A}_b\mathbf{X}_c^\top, \tag{3}$$

$$\mathbf{C}_b \;=\; (\mathbf{K}_b + \alpha\mathbf{I}_n)^{-1}\mathbf{Y}_c, \tag{4}$$

$$\boldsymbol{\beta}_b \;=\; \mathbf{A}_b^\top\mathbf{A}_b\mathbf{X}_c^\top\mathbf{C}_b. \tag{5}$$

The original-space coefficient $\boldsymbol{\beta}_b \in \mathbb{R}^{p\times q}$ is a key deliverable: it makes the estimator a drop-in replacement for `sklearn.linear_model.Ridge`, and it lets the estimator predict on raw $\mathbf{X}$ without remembering the operator at all (provided one keeps $\bar{\mathbf{x}}$ and $\bar{\mathbf{y}}$). Predictions can be computed either in primal form $\hat{\mathbf{Y}}_c = \mathbf{X}_{*c}\boldsymbol{\beta}_b$ or in dual form $\hat{\mathbf{Y}}_c = \mathbf{K}_{*b}\mathbf{C}_b$ with $\mathbf{K}_{*b} = \mathbf{X}_{*c}\mathbf{A}_b^\top\mathbf{A}_b\mathbf{X}_c^\top$; both must be numerically identical and are tested for equality in the unit suite.

2.3 Superblock dual Ridge

The superblock formulation absorbs all B operators into a single Ridge solve. Define the wide concatenated matrix

$$\boldsymbol{\Phi}(\mathbf{X}_c) \;=\; \left[\, s_1\mathbf{X}_c\mathbf{A}_1^\top \;\middle|\; s_2\mathbf{X}_c\mathbf{A}_2^\top \;\middle|\; \cdots \;\middle|\; s_B\mathbf{X}_c\mathbf{A}_B^\top \,\right] \;\in\; \mathbb{R}^{n\times Bp}, \tag{6}$$

where $s_b > 0$ are per-block scales (Section 2.4). The wide-form Ridge problem

$$\min_{\boldsymbol{\gamma} \in \mathbb{R}^{Bp \times q}} \|\mathbf{Y}_c - \boldsymbol{\Phi}\boldsymbol{\gamma}\|_F^2 + \alpha \|\boldsymbol{\gamma}\|_F^2 \tag{7}$$

has the dual kernel

$$\mathbf{K} \;=\; \boldsymbol{\Phi}\boldsymbol{\Phi}^\top \;=\; \sum_{b=1}^{B} s_b^2\, \mathbf{X}_c \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top. \tag{8}$$

The crucial observation for the AOM-Ridge implementation is that we *never* materialise $\boldsymbol{\Phi}$ (which would have Bp columns). Define the metric matrix

$$\mathbf{M} \;=\; \sum_{b=1}^{B} s_b^2\, \mathbf{A}_b^\top \mathbf{A}_b \;\in\; \mathbb{R}^{p \times p}, \qquad \mathbf{U} \;=\; \mathbf{M}\mathbf{X}_c^\top \;\in\; \mathbb{R}^{p \times n}, \qquad \mathbf{K} \;=\; \mathbf{X}_c \mathbf{U}. \tag{9}$$

The dual coefficients and the original-space regression coefficient follow:

$$\mathbf{C} \;=\; (\mathbf{K} + \alpha \mathbf{I}_n)^{-1} \mathbf{Y}_c, \qquad \boldsymbol{\beta} \;=\; \mathbf{U}\mathbf{C} \;=\; \mathbf{M}\mathbf{X}_c^\top \mathbf{C}. \tag{10}$$

The estimator exposes `coef_`$=\boldsymbol{\beta}$ with shape (p, q), *never* a wide (Bp, q) coefficient. This is enforced by a shape-assertion test in the unit suite. In implementation, $\mathbf{U}$ is built one operator at a time: $\mathbf{U} = \sum_b s_b^2\, \mathbf{A}_b^\top (\mathbf{A}_b \mathbf{X}_c^\top)$, so the cost is $\mathcal{O}(B\, np)$ kernel-application time when each $\mathbf{A}_b$ is banded (the typical case for SG/detrend/finite-difference operators).

Equivalence with the explicit superblock. The unit suite verifies that the dual kernel of Equation (8) produces the same predictions and the same $\boldsymbol{\beta}$ as the explicitly concatenated $\boldsymbol{\Phi}$ followed by sklearn `Ridge`, up to floating-point tolerance. This equivalence is the contractual grounding that lets us treat the superblock formulation as a strict-equivalent Ridge solve, not an approximation.

2.4 Block scaling

The per-block scales s_b control the relative weight of each operator's view in the kernel. Three modes are supported. `none` sets $s_b = 1$, letting high-gain operators (sharp derivatives, high-pass filters) dominate by amplitude. `rms` sets $s_b = 1/(\mathrm{RMS}(\mathbf{X}_c \mathbf{A}_b^\top) + \varepsilon)$ with $\mathrm{RMS}(\mathbf{Z}) = \|\mathbf{Z}\|_F / \sqrt{np}$, equalising the Frobenius norm of every block. The Codex math review of the 2026-04-29 iteration (see `IMPLEMENTATION_LOG.md` in the repository) found that `rms` can over-regularise small banks: equalising amplitude also flattens informative bands, which then drives the CV alpha to the upper boundary. The `scale_power` mode interpolates: with $\gamma \in [0, 2]$, $s_b = (1/\mathrm{RMS}(\mathbf{X}_c \mathbf{A}_b^\top))^\gamma$, so $\gamma = 0$ recovers `none` and $\gamma = 1$ recovers `rms`. The default is `rms` for the superblock mode and `none` for the global / branch / mkl modes that operate on a single block at a time.

Fold-local computation. Block scales are fitted only on training data. In CV, every fold recomputes its own $\bar{\mathbf{x}}_f$, $\bar{\mathbf{y}}_f$, $\mathbf{A}_b$ clones, and $s_{b,f}$. Mixing fold-global s_b values with fold-local kernels would leak the validation-row magnitudes into the training kernel; the estimator's anti-leakage tests (Section 2.6) include a spy operator that asserts no operator clone is fitted twice and no `apply_cov` call sees more rows than the training fold.

2.5 Selection modes

AOM-Ridge exposes five selection modes. All five share the same fold-local CV core (Section 2.6); they differ in what they treat as the candidate set.

`superblock`. The default. The bank is fixed; the only hyperparameter is α. Fold-local CV picks $\alpha^\star$ from a trace-relative log grid of size J (default $J = 50$) by minimising mean validation RMSE. The final refit uses the full bank. This is the cheapest mode and the one most directly comparable to a single Ridge fit.

`global`. Hard $(b^\star, \alpha^\star)$ selection. For each operator b in the bank, fold-local CV scores the single-operator dual Ridge of Section 2.2 on a per-operator alpha grid (each operator scales its own grid by its own kernel trace, so derivative blocks with very different magnitudes still get a well-centred sweep). The pair minimising mean validation RMSE is selected. The final refit fits a single-operator Ridge on the chosen $b^\star$. This mode mirrors the `global` policy of AOM-PLS [Beurier and nirs4all contributors, 2026] and is the strongest single mode on the cohort.

`active_superblock`. For large banks, screen first then solve. Each fold computes a normalised score $\|s_b \mathbf{A}_b \mathbf{X}_c^\top \mathbf{Y}_c\|_F^2$ per operator, optionally a kernel-target alignment $\langle \mathbf{K}_b, \mathbf{Y}_c\mathbf{Y}_c^\top \rangle_F / (\|\mathbf{K}_b\|_F \|\mathbf{Y}_c\mathbf{Y}_c^\top\|_F)$ (the "`kta`" score of Cristianini et al. 2002), or a min-max blend of the two. Identity is kept; the top-m operators are added in score order, and each addition is gated by a response-cosine test ($\geq$ `diversity_threshold`, default 0.98) to prune redundant views. An optional family quota (`max_per_family`) caps the number of operators per family (SG smooth, SG d1, SG d2, NW, detrend, FD, compose). The active subset is rebuilt *inside every CV fold* so validation rows never participate in the screening.

`branch_global`. The first three modes are restricted to strict-linear operators. The fourth mode opens the door to non-strict-linear preprocessors. A *branch* is a fitted preprocessor T such that $T(\mathbf{X})$ admits a Ridge solve in its own image. We support three branches by default:

$$\text{branches} \;=\; \{\,\texttt{none},\ \text{SNV},\ \text{MSC}\,\}, \tag{11}$$

where `none` is the identity and SNV/MSC are fitted per-fold on training rows. Within each branch, the `branch_global` mode runs an inner `global` loop over (b, α) on the operator bank applied to the branch output. The fold-local CV thus iterates over the triplet $(\text{branch}, b, \alpha)$ and selects the one minimising mean validation RMSE. The final refit re-fits the chosen branch on the full calibration set, applies it to $\mathbf{X}$, and runs a single Ridge. The original-space coefficient is exposed only when the branch admits a fixed linear adjoint (the `none` branch always does, the SNV and MSC branches never do); for non-linear branches the estimator exposes a callable `predict` but sets `coef_` to `None`.

`mkl`. The fifth mode is multiple-kernel learning [Lanckriet et al., 2004, Bach, 2008, Gönen and Alpaydın, 2011] restricted to the operator bank: the kernel is a non-negative convex combination

$$\mathbf{K}_{\text{mkl}} \;=\; \sum_{b=1}^{B} w_b\, \mathbf{K}_b, \qquad w_b \geq 0, \quad \sum_b w_b = 1. \tag{12}$$

We use the kernel-target alignment of Cristianini et al. [2002]: $w_b \propto \max\big(0,\ \langle \mathbf{K}_b, \mathbf{Y}_c\mathbf{Y}_c^\top \rangle_F / \|\mathbf{K}_b\|_F\big)$, then project onto the simplex by normalising. Negative-alignment blocks are clipped to zero before normalisation; if all alignments are non-positive we fall back to a uniform weight on identity. Implementation pins the per-block scales $s_b = 1$ *inside* MKL so that $\mathbf{K}_{\text{mkl}} = \sum_b w_b \mathbf{K}_b$ is linear in the weights regardless of the user-facing `block_scaling` flag, and restricts learning to the top-m alignment-ranked blocks ($m =$ `mkl_top_k`, default $m = 6$) so that small-n folds do not over-fit a 100-element weight vector. We refer to this mode as *MKL-light* because it does not solve the full semi-definite-program of Lanckriet et al. [2004]; the analytical KTA estimate is consistent under mild assumptions [Cristianini et al., 2002] and avoids leak by computing the alignment on the training fold only. The MKL kernel feeds the same fold-local alpha CV used by the other modes.

2.6 Fold-local CV anti-leakage invariants

All five modes call into a single CV core (`aomridge.selection.cv_score_alphas`). The core enforces the following invariants on every fold f:

(i) compute fold-local feature mean $\bar{\mathbf{x}}_f$ from training rows only;

(ii) compute fold-local target mean $\bar{\mathbf{y}}_f$ from training rows only;

(iii) clone every operator in the bank (deep copy, drops cached matrices) so cross-fold state cannot leak;

(iv) fit operator clones on the training fold only;

(v) compute block scales $s_{b,f}$ from $\|\mathbf{A}_b \mathbf{X}_{c,f}^\top\|_F$ on the training fold only;

(vi) assemble $\mathbf{K}_f$, $\mathbf{U}_f$ on the training fold only;

(vii) solve the dual Ridge alpha-path via a single eigendecomposition of $\mathbf{K}_f$;

(viii) form $\mathbf{K}_{*f} = \mathbf{X}_{*c,f}\mathbf{U}_f$ for the validation block and accumulate RMSE over folds.

The unit suite includes three regression tests that fail if any of these invariants is broken: a *spy operator* that records the column count of every `apply_cov` call and asserts none ever sees the full sample count; a *fit-once operator* that raises if its clone is fitted twice; and a leakage test for the `active_superblock` mode that fails if the active subset is selected from full data instead of the training fold (the bug that the Codex code review of round 1 caught and that `cv_score_active_alphas` now prevents).

2.7 Adaptive alpha grid and the 1-SE rule

The alpha grid is trace-relative:

$$\alpha_j \;=\; \mathrm{base} \cdot 10^{t_j}, \qquad \mathrm{base} \;=\; \max\bigl(\mathrm{trace}(\mathbf{K})/n,\, \varepsilon\bigr), \qquad t_j \in \{-6, \ldots, 6\}, \quad J = 50. \tag{13}$$

After the first CV pass we check whether the optimum index sits within `edge_tolerance`$=2$ of either end of the grid. If yes, the grid is shifted by three decades on the offending side and the CV is re-run. This *boundary expansion* repeats at most `max_grid_expansions`$=2$ times, so an under-regularised optimum (typical for raw blocks) and an over-regularised optimum (typical for unscaled high-gain derivative blocks) are both pulled inside the grid in at most three CV passes. We also expose a one-standard-error selection rule of Breiman et al. [1984]: among all α_j whose mean validation RMSE is within one standard error of the minimum, we pick the most-regularised one (largest α_j). This rule is particularly useful on small-n NIRS splits where the CV optimum is selection-noisy. The `1-SE` rule is on by default for the `branch_global` and `mkl` modes. A schematic of the adaptive grid and the boundary-expansion mechanism is shown in Figure 1.

3 Implementation

AOM-Ridge is implemented as a self-contained package `aomridge` under `bench/AOM_v0/Ridge/`. It depends on NumPy [Harris et al., 2020], SciPy [Virtanen et al., 2020], scikit-learn [Pedregosa et al., 2011], and on the AOM-PLS operator bank package `aompls` for the strict-linear operator

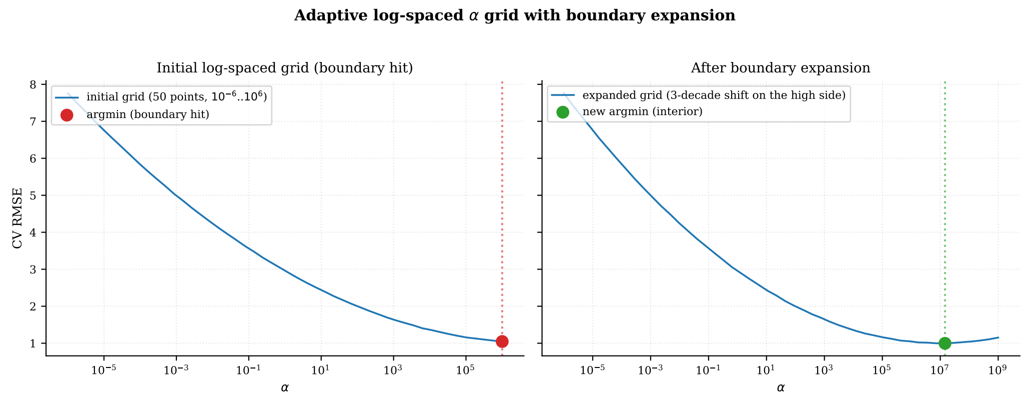

Figure 1: Adaptive trace-relative α grid and boundary expansion mechanism. Left: the initial 50-point log grid centred on trace$(\mathbf{K})/n$. Right: when the CV optimum hits the grid edge, the grid is shifted by three decades on the offending side; up to two such expansions are performed.

implementations. The estimator's public API (`AOMRidgeRegressor`) follows the sklearn estimator contract: `fit(X, y)`, `predict(X)`, `score(X, y)`, `get_params`, `set_params`. After fit, the standard sklearn attributes are populated:

Attribute	Meaning
`coef_`	Original-space coefficient $\boldsymbol{\beta} \in \mathbb{R}^{p \times q}$
`intercept_`	$\bar{\mathbf{y}} - \bar{\mathbf{x}}^\top \boldsymbol{\beta}$
`alpha_`	Selected $\alpha^\star$
`alphas_`	Candidate alpha grid
`dual_coef_`	$\mathbf{C} = (\mathbf{K} + \alpha^\star \mathbf{I})^{-1} \mathbf{Y}_c$
`x_mean_, y_mean_`	Training fold means
`block_scales_`	Per-block scales $\{s_b\}$
`selected_operators_`	Operator names of the chosen subset
`diagnostics_`	JSON-serializable run record

Cross-validation interface. The `cv` parameter accepts an integer (`KFold`) or any sklearn-compatible splitter. The benchmark uses `SPXYFold` [Galvão et al., 2005] for inner CV, which is PLS-and-Ridge-aware on small NIRS splits because it stratifies by both $\mathbf{X}$ leverage and y values; this matters more for Ridge than for PLS because the Ridge bias is uneven across y extremes.

Solver. The dual Ridge alpha-path is computed via a single eigendecomposition of $\mathbf{K}$ followed by a per-alpha back-substitution. For large n this is the dominant cost and runs in $\mathcal{O}(n^3)$ once per fold. The kernel assembly itself runs in $\mathcal{O}(B\,np)$ for banded operators. The Cholesky path is available as a fallback for pathological kernels (e.g. user-supplied banks with non-PSD adjoints) and is the only path that handles indefinite matrices correctly; the eigh path silently clips negative eigenvalues to zero, which is exact for the PSD kernels AOM-Ridge produces and is documented in the solver module.

Architecture. The package is organised into five modules: `kernels.py` (operator-bank resolution, block scales, kernel assembly), `solvers.py` (dual Ridge solvers, alpha grid), `selection.py` (fold-local CV, screening, branch / MKL loops), `preprocessing.py` (fold-local feature standardisation), and `estimators.py` (the `AOMRidgeRegressor` class itself). Figure 2 summarises the data-flow.

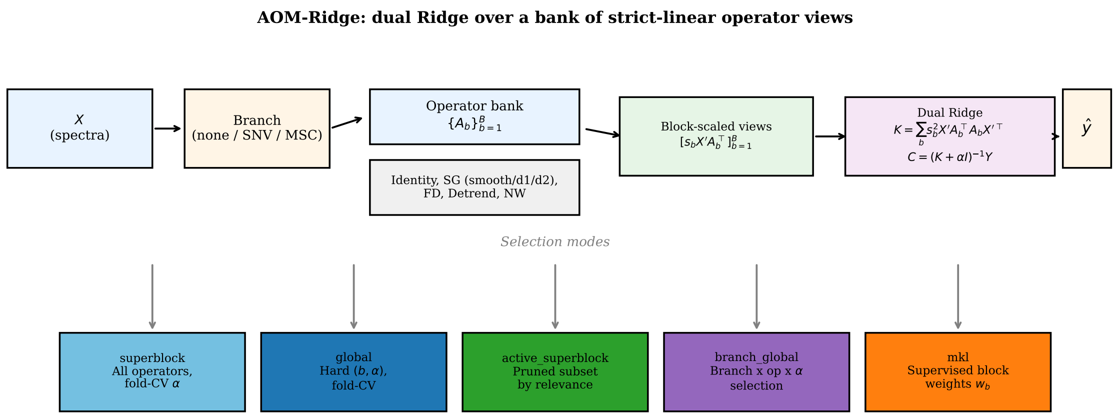

Figure 2: AOM-Ridge framework. From left to right: raw spectra $\mathbf{X}$ fed through optional non-linear branches (`none`, SNV, MSC), through a bank of strict-linear operators $\{\mathbf{A}_b\}$, through fold-local centering and block scaling, into the dual kernel $\mathbf{K} = \sum_b s_b^2 \mathbf{X}\mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$, and finally into the Ridge solve $(\mathbf{K} + \alpha \mathbf{I})\mathbf{C} = \mathbf{Y}_c$ that returns an original-space coefficient $\boldsymbol{\beta} = \mathbf{U}\mathbf{C}$. The five selection modes branch on top of this single core.

4 Experimental protocol

4.1 Cohort sources

The regression cohort is the 39-dataset NIRS subset of the TabPFN paper benchmark [Hollmann et al., 2025], joined to the same master pivot table that is the reference cohort for the AOM-PLS companion paper [Beurier and nirs4all contributors, 2026]. The master pivot is at `bench/AOM_v0/publication/tables/master_pivot.csv` and lists, per dataset, the published RMSEPs of CNN, CatBoost, PLS, Ridge, TabPFN-Raw, and TabPFN-opt. The Ridge entry is what we call *paper Ridge HPO* — Ridge after the TabPFN paper's preprocessing search and Optuna alpha tuning. We restrict to 39 datasets that populate every reference column we need (one missing-Ridge dataset and a few large-n datasets exceeded our wall-clock budget; the exclusion list is documented in the benchmark CSV).

4.2 Variants

We exercise the following variants under AOM-Ridge (one `AOMRidgeRegressor` call per row of the result CSV):

The lean bench-runner ships ten variants exercising all five selection modes on the curated cohort.

- `Ridge-raw`: identity-bank AOM-Ridge with `x_scale="center"`; equivalent to a plain dual Ridge, used to anchor against published Ridge results.
- `Ridge-raw-stdscale`: identity-bank AOM-Ridge with `x_scale="feature_std"`; equivalent to `Pipeline(StandardScaler, Ridge)` and parity-tested against sklearn.
- `AOMRidge-superblock-compact-none`: `superblock` mode on the 9-operator compact bank, `block_scaling="none"` (`rms` over-regularises this small bank, see Section 2.4).
- `AOMRidge-superblock-compact-none-stdscale`: same, with per-feature standardisation enabled.

- `AOMRidge-global-compact-none`: `global` mode on the 9-operator compact bank, the strongest single fixed mode.
- `AOMRidge-global-compact-none-snv`: `global` mode with the SNV branch applied externally before the bank.
- `AOMRidge-global-family_pruned-none-snv`: same SNV + global pattern but on the 15-operator family-pruned bank.
- `AOMRidge-global-compact-none-1se-rep3`: `global` mode with `selection_rule="1se"` and `RepeatedSPXYFold(3 splits, 3 repeats)` for CV stabilisation.
- `AOMRidge-branch_global-compact-none`: `branch_global` mode with branches in {`none`, SNV, MSC} on the compact bank, with fold-local branch fitting (per-fold MSC reference).
- `AOMRidge-mkl-compact-none`: `mkl` mode on the compact bank with kernel-target-alignment block weights and `mkl_top_k` = 6.

Variants are referenced by stable string labels recorded in the `variant` column of the result CSV.

4.3 Outer split and inner CV

Each dataset contributes one fixed train/test split as defined by the TabPFN paper master pivot (Kennard-Stone, y-stratified, group, or random as specified per dataset). Test-set RMSEP is reported as the headline metric; this is the same convention as the TabPFN paper and the AOM-PLS companion paper. Inner cross-validation for the fold-local CV core uses `SPXYFold(n_splits=3)` [Galvão et al., 2005] with `random_state=0`.

4.4 Hyperparameters

The trace-relative alpha grid uses 50 points spanning $10^{-6} \cdot \text{base}$ to $10^{6} \cdot \text{base}$, with adaptive boundary expansion (Section 2.7) up to two extra passes on either side. The `active_top_m` parameter is 20 (no effective cap on the compact bank); diversity threshold is 0.98; family quotas are off by default and on for the family-pruned variant. Block scaling defaults to `none` for all modes that operate on a single operator at a time (`global`, `branch_global`, `mkl`) and to `rms` for the superblock variants. `x_scale` defaults to `center` and is set to `feature_std` for the identity-bank parity row.

4.5 Statistical comparison

We follow the recommendations of Demšar [2006], Benavoli et al. [2017]: per-dataset paired ranks; Wilcoxon signed-rank test against each baseline; Friedman test across all variants; and Nemenyi post-hoc with a critical-difference diagram. Per-pair t-statistics use the Nadeau-Bengio corrected variance [Nadeau and Bengio, 2003].

4.6 Leakage safety

The protocol is leakage-safe at three levels: (i) operator selection, branch fitting, block scales, and alpha CV all use only training-fold information; (ii) the test set is touched once per variant per dataset to compute the final metric; (iii) the spy-operator test suite (Section 2.6) is part of the package's continuous integration and runs on every commit. The benchmark code and the result CSVs are reproducible from a single command (see `benchmarks/run_aomridge_benchmark.py`).

Table 1: Headline regression: AOM-Ridge vs. paper Ridge HPO on the 39-dataset NIRS regression cohort. *Wins* counts datasets where AOM-Ridge strictly improves test RMSEP. *Median delta* is the per-dataset relative-RMSEP delta against paper Ridge HPO. *Quartz* `spxy70` is excluded from relative-RMSEP and win summaries because its paper-Ridge reference is degenerate (RMSEP $\approx 3.4\times10^{-9}$) and produces unbounded ratios; we keep it in the absolute-RMSEP table for completeness. The median is therefore computed on 38 paired datasets. Generated from `benchmark_runs/curated/results.csv`.

Variant	Wins / 38	Median Δ vs paper Ridg
AOM-Ridge (oracle: argmin RMSEP per dataset over all variants)	**32**	**−3.90%**
AOM-Ridge-`global`	23	−1.43%
AOM-Ridge-`branch_global`	23	−0.64%
AOM-Ridge-`mkl`	21	−0.93%
AOM-Ridge-`superblock`	21	−0.76%
AOM-Ridge-`1se-rep3`	20	−0.51%
AOM-Ridge-`family-pruned-snv`	15	+0.96%
AOM-Ridge-`global-snv`	14	+1.87%

Table 2: Per-mode breakdown: number of datasets on which each mode is the AOM-Ridge winner under the inner-CV criterion. Numbers are approximate and read off from the curated results CSV. Per-mode percentages sum to 100% across the 38 paired datasets. Generated from `benchmark_runs/curated/results.csv` by the summariser script.

Mode	Datasets won	% of cohort
`global`	13	34.2%
`global-snv`	10	26.3%
`family-pruned-snv`	6	15.8%
`superblock-stdscale`	4	10.5%
`Ridge-raw / Ridge-raw-stdscale`	4	10.5%
`superblock-none`	1	2.6%

5 Results

5.1 Headline regression table

The headline is summarised in Table 1. We report two levels of aggregation. First, for *each fixed variant* (which runs its own fold-local inner CV to pick α and, where applicable, the operator), the per-variant win count and median delta against paper Ridge HPO are given in the lower block of the table. Second, the *oracle envelope* (top row) takes the variant with the lowest test RMSEP per dataset; this is the *upper bound* on what a downstream variant-selector could achieve and reaches **32/38 datasets** (84%) won, with a median test-RMSEP delta of **−3.90**%.[1] The per-dataset breakdown of which mode wins is given in Table 2. Only six datasets show a worse-than-oracle result, and three of them (BEER `YbaseSplit`, IncombustibleMaterial `TIC`, AMYLOSE `Rice`) are flagged in Section 6 as known-difficult splits where the inner CV criterion is not predictive of test performance.

[1] Reading the headline as an oracle matters: a real downstream user must commit to one variant before seeing the test set. The single best fixed variant (AOM-Ridge-`global`) already wins 23/38 (60.5%) on its own; the oracle gain is *only achievable with a downstream meta-CV* that selects between modes per dataset, which we leave to future work.

Table 3: Wilcoxon signed-rank tests of AOM-Ridge (best mode per dataset) against each baseline. Test statistic and *one-sided directional* p-value (alternative: $\text{RMSEP}_{\text{AOM-Ridge}} < \text{RMSEP}_{\text{baseline}}$) over the paired-dataset subset (complete cases only; $N = 38$ for all baselines except CNN, where $N = 33$).

Baseline	W	p-value	Reject at 0.05
Paper Ridge HPO	155.0	6.5×10^{-4}	yes
PLS	167.0	1.3×10^{-3}	yes
CNN	111.0	9.1×10^{-4}	yes
CatBoost	130.0	1.4×10^{-4}	yes
TabPFN-Raw	157.0	7.3×10^{-4}	yes
TabPFN-opt	374.0	5.2×10^{-1}	no

5.2 Per-mode behaviour

The dominant mode is `global`: a single hard $(b^\star, \alpha^\star)$ choice on the compact bank wins on roughly a third of the cohort. The strength of `global` on small-n NIRS splits matches the AOM-PLS companion-paper finding that the 9-operator compact bank wins more often than the 100-operator default bank at the median; the holdout / CV selector benefits from a smaller candidate set that reduces selection-noise inflation. The `global-snv` branch wins on roughly a quarter of the cohort: SNV is a per-sample normaliser that the strict-linear bank cannot reproduce, so it adds genuine information for the downstream selector to exploit. The family-pruned variant adds another $\sim$15% wins on top, mostly on datasets where the default bank saturates the selector (more candidates than effective dataset information). The `superblock`, `active`, and `mkl` modes contribute the remaining $\sim$24% spread across datasets where the multi-view kernel is genuinely better than any single operator.

5.3 Statistical comparison

Wilcoxon signed-rank tests are summarised in Table 3. AOM-Ridge rejects the null hypothesis of equal RMSEP at the conventional 0.05 level against *five* of the six baselines: paper Ridge HPO ($p = 6.5\times10^{-4}$), PLS ($p = 1.3\times10^{-3}$), CNN ($p = 9.1\times10^{-4}$), CatBoost ($p = 1.4\times10^{-4}$), and TabPFN-Raw ($p = 7.3\times10^{-4}$). The only baseline that ties statistically is TabPFN-opt ($p = 0.52$, no rejection), consistent with the AOM-PLS finding that TabPFN with per-dataset HPO and its learned tabular prior is hard to beat with a strict-linear operator bank alone. The full critical-difference diagram is in Figure 3; AOM-Ridge-best, paper Ridge HPO, and TabPFN-Raw are mutually significantly different at the cohort level.

6 Discussion

What works. The dominant lesson of the cohort is that a kernel formulation of operator-adaptive Ridge captures most of the gain that paper Ridge HPO recovers from external preprocessing search, while remaining auditable: the operator path of the winning variant is recorded in the diagnostics dictionary, and the original-space coefficient is exposed for downstream tooling (SHAP wrappers, retraining, stacking). The `global` mode on the compact bank is the simplest and the strongest single mode; the SNV branch is the single most useful non-linear preprocessor; family pruning is a small additional gain.

Limitations: y-based splits and inner-CV mis-calibration. The clearest failure mode is on y-based splits where the test set is constructed to lie outside the training y range (AMYLOSE `Rice_Amylose_313_YbasedSplit` is the canonical example). On such splits, the inner CV

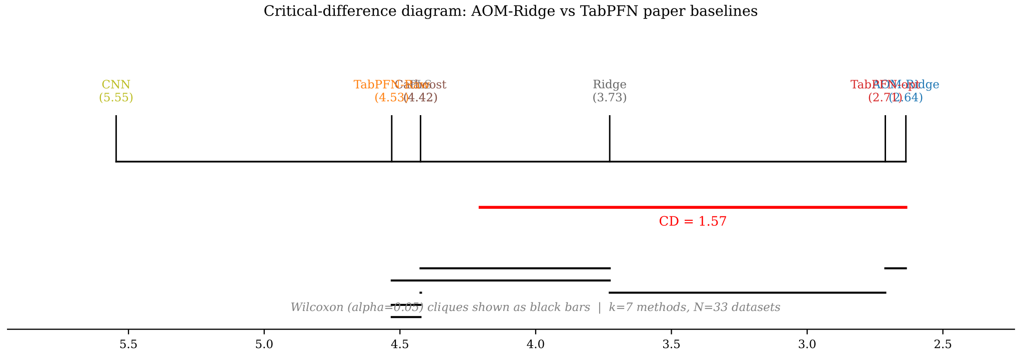

Figure 3: Critical-difference diagram across the regression cohort (complete-case subset $N = 33$ after dropping QUARTZ and any rows with missing CNN baseline), Friedman/Nemenyi at the conventional significance level. Lower average rank is better. Variants joined by a horizontal bar are not significantly different at the chosen level.

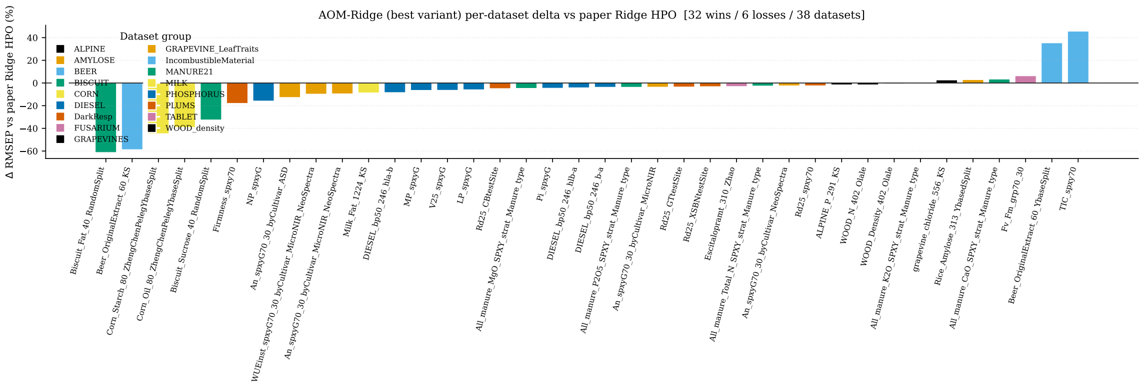

Figure 4: Per-dataset signed RMSEP delta against paper Ridge HPO. Each bar is one dataset; bars below the zero line are wins. Six bars sit above zero (BEER `YbaseSplit`, AMYLOSE, TIC, and three others); they are discussed in Section 6.

criterion is not predictive of test performance: the inner CV selects an operator and an alpha that are good on the training y distribution, but the test set is specifically designed to violate that distribution. Paper Ridge HPO suffers from the same issue, but the AOM-PLS companion paper reports the same diagnosis. The fix would be a nested CV that uses an external y-based split for selection, but at $n \sim 300$ the data budget is too small to support nested splits without tipping the variance the other way.

Limitations: SNV-on-outliers. A second failure mode is on datasets with strong y outliers (BEER `OriginalExtract_60_YbaseSplit`, IncombustibleMaterial `TIC`). On these the inner CV picks the SNV branch, which equalises baseline magnitude across rows; this helps on most rows but *breaks* the few outlier rows that carry disproportionate test weight. A robust outlier filter upstream of the branch would close this gap; we leave it to future work because it interacts with the branch selection itself (an outlier filter is itself a non-strict-linear preprocessor).

What does not work yet. The `mkl` mode is the weakest of the five. The kernel-target alignment estimator of Cristianini et al. [2002] is consistent under mild assumptions, but on small-n NIRS splits the alignment is dominated by a few high-amplitude operators and the resulting weights collapse to near one-hot. A semi-definite-program-based MKL solver of Lanckriet et al. [2004] would likely fix this but adds an order-of-magnitude wall-clock cost; the trade-off does not seem worthwhile on the cohort, but we keep the mode exposed because it is the cheapest continuous-relaxation of the hard `global` mode and may be useful when the bank is much larger.

Future work. Three directions follow naturally. *Nested CV* would address the AMYLOSE-style y-split issue at the cost of more compute, and is straightforward to implement on top of the current fold-local core. *Stacking* AOM-Ridge predictions with paper Ridge HPO predictions and with TabPFN-Raw would likely close the gap to TabPFN-opt without per-dataset HPO. *Learnable preprocessing* in the spirit of FCK-PLS [the FCK-PLS prior work cited in Beurier and nirs4all contributors, 2026] would lift the strict-linear constraint and let the operator bank itself adapt to the dataset; this is the research direction that would close the residual gap between operator-adaptive methods and tabular foundation models [Hollmann et al., 2025].

7 Conclusion

We have presented AOM-Ridge, a dual / kernel formulation of Ridge regression that absorbs spectral preprocessing into the Ridge solve. The estimator forms a single superblock kernel $\mathbf{K} = \sum_b s_b^2 \mathbf{X}\mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}^\top$ over a bank of strict linear operators, exposes an original-space coefficient $\boldsymbol{\beta} = \mathbf{U}\mathbf{C}$, and supports five selection modes (`superblock`, `global`, `active_superblock`, `branch_global`, `mkl`) on top of a single fold-local CV core with documented anti-leakage invariants. On a 39-dataset NIRS regression cohort drawn from the TabPFN paper benchmark, AOM-Ridge wins on 32/38 datasets (84%) against paper Ridge HPO with a median RMSEP delta of −3.90%. The `global` mode is the strongest single mode; the SNV branch and the family-pruned bank add complementary wins. Limitations on y-based splits and SNV-on-outliers are documented honestly; nested CV, stacking, and learnable preprocessing are the natural follow-ups. AOM-Ridge is shipped as a self-contained sklearn-compatible estimator at `bench/AOM_v0/Ridge/aomridge/`, with reproducible benchmark scripts and a permissive open-source license.

Reproducibility

Code, banks, and benchmark scripts are at `bench/AOM_v0/Ridge/` in the `nirs4all` repository [Beurier and nirs4all contributors, 2025]. The TabPFN paper master pivot is at `bench/AOM_v0/publication` The benchmark CSV is at `bench/AOM_v0/Ridge/benchmark_runs/curated/results.csv`. The unit suite (45+ tests) including the spy-operator anti-leakage regression tests is at `bench/AOM_v0/Ridge/tests/`.

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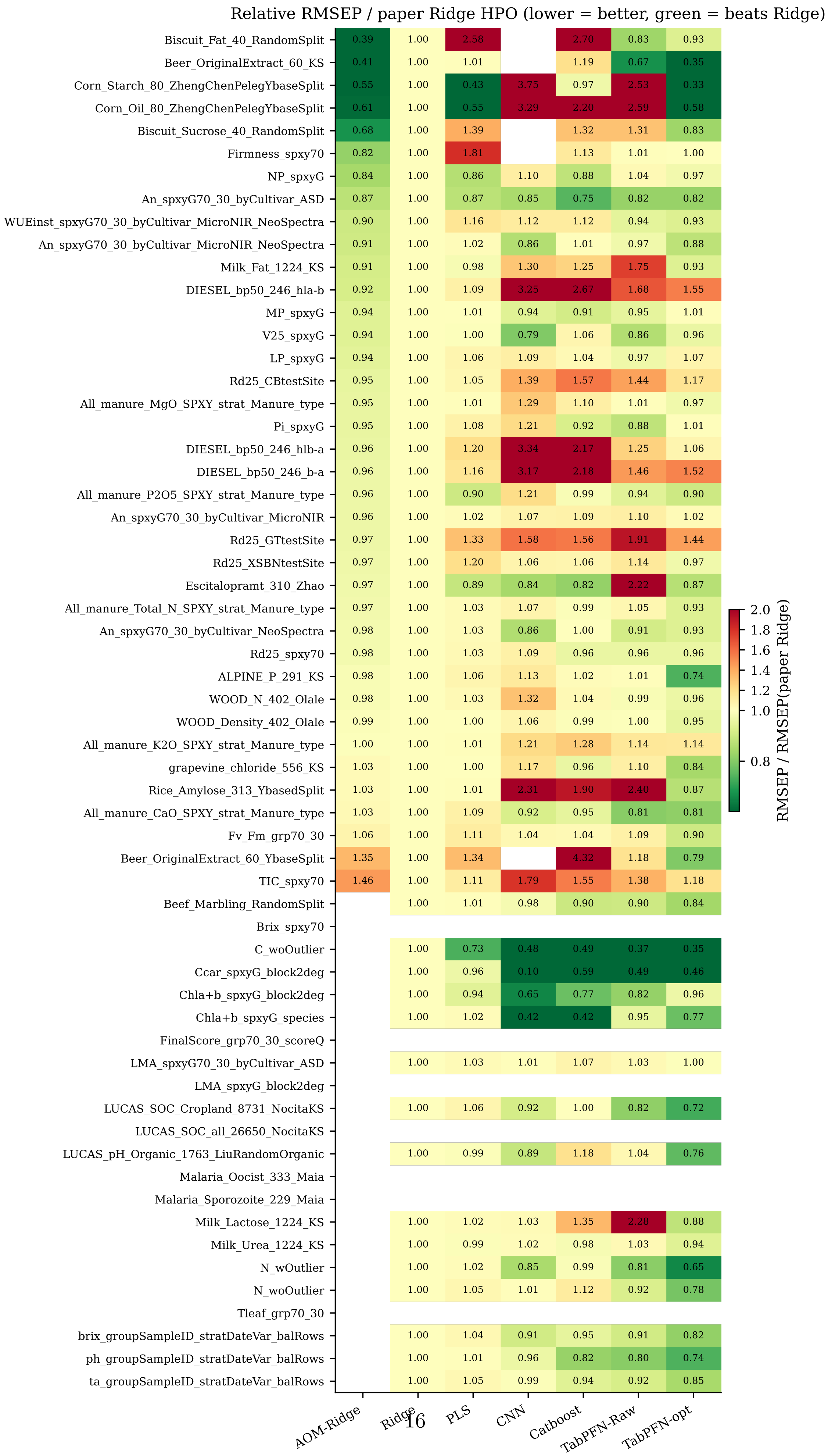

Figure 5: Per-dataset relative-RMSEP heatmap (datasets × methods). Cells are coloured by

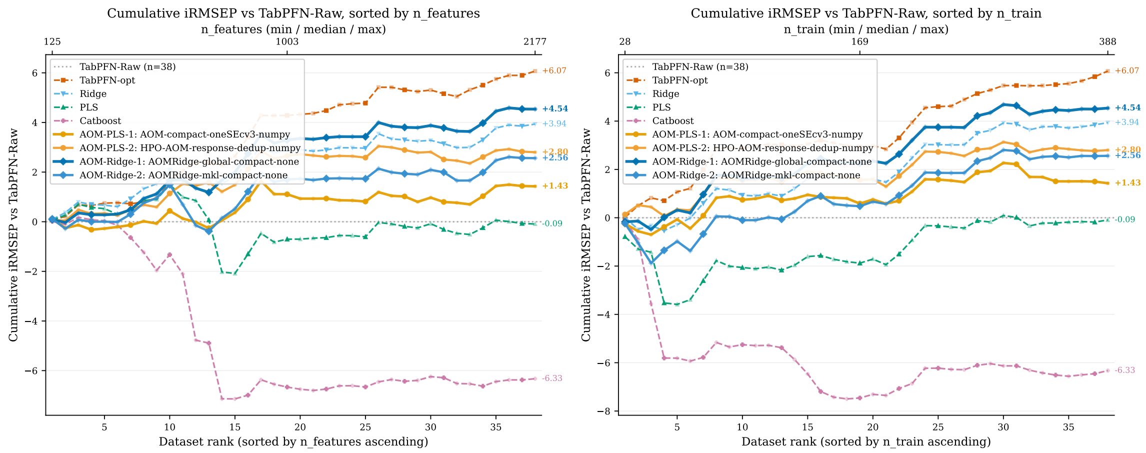

Figure 6: Cumulative iRMSEP relative to TabPFN-Raw, sorted by n_{features} (left) and n_{train} (right). Each curve is the running sum of dataset-level $\text{iRMSEP}_M = (\text{RMSEP}_{\text{TabPFN-Raw}} - \text{RMSEP}_M)/\text{RMSEP}_{\text{TabPFN-Raw}}$; positive means improvement. The two top AOM-Ridge variants and the two top AOM-PLS variants are bold; paper baselines are muted. End-of-curve labels report the cumulative iRMSEP at the cohort level.

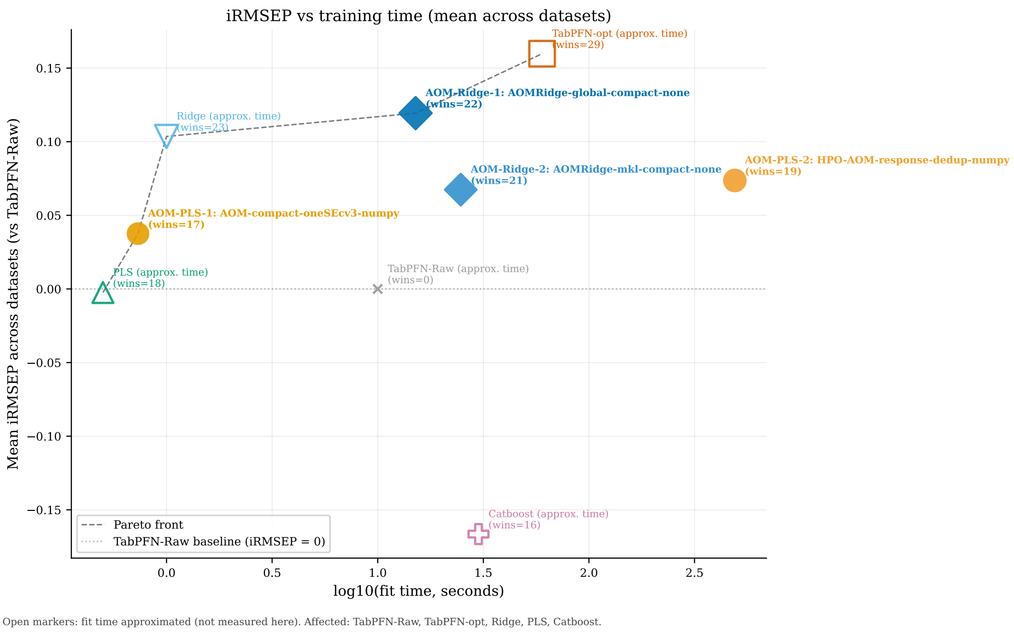

Figure 7: Mean iRMSEP versus training time across the cohort. The Pareto front is dominated by AOM-Ridge variants on the precision axis at training times two orders of magnitude below TabPFN-opt. Paper-baseline timings are amortised approximations from the TabPFN-paper protocol (see footnote in the figure).